

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I D. Gnana Deepthi (192411123) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D. Gnana Deepthi

APPLICATION- BASED QUESTIONS

3. Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- Demonstrate how the encryption process transforms the message step-by step.
- Explain how the repeating keyword influences each character of the ciphertext.

A company uses a keyword-based encryption technique (Vigenère Cipher) to encrypt the word "HELLO" using the keyword "KEY".

a. Encryption Process (Step by Step)

Plaintext: HELLO

Keyword: KEY

Repeat the keyword until it matches the plaintext length.

Plaintext: H E L L O

Keyword: K E Y K E

Convert letters to numbers (A = 0, B = 1, ..., Z = 25).

Letter : H E L L O

Value : 7 4 11 11 14

Keyword : K E Y K E

Value: 10 4 24 10 4

Use the formula:

Ciphertext = (Plaintext + Keyword) mod 26

Plaintext: 7 4 11 11 14

Keyword: 10 4 24 10 4

Sum: 17 8 35 21 18

Mod 26: 17 8 9 21 18

Convert back to letters:

17 = R

8 = I

9 = J

21 = V

18 = S

Ciphertext = RIJVS

b. How the Repeating Keyword Influences the Ciphertext

- The keyword **KEY** is repeated as **KEYKE**.
- Each letter of the keyword shifts the corresponding plaintext letter by a different amount.
- Different shifts make the same plaintext letter encrypt differently depending on its position.
- This makes frequency analysis much harder than in the Caesar cipher, providing better security.

4. Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyse how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Investigators suspect that a company logo contains hidden confidential data.

a. Detecting Hidden Data in the Image

Investigators can detect hidden data using the following methods:

1. **LSB (Least Significant Bit) Analysis**
 - Examine the least significant bits of image pixels.
 - Unusual patterns may indicate hidden information.
2. **Statistical Analysis**
 - Compare pixel distributions with those of a normal image.
 - Hidden data often changes image statistics.
3. **Steganalysis Tools**
 - Use forensic software such as **StegExpose**, **Stegdetect**, or **OpenStego** to identify hidden messages.
4. **Metadata Examination**

- Check image metadata for unusual modifications or embedded information.

b. Two Techniques to Improve Security of Hidden Communication

1. Encrypt the Secret Data Before Hiding

- Encrypt the message using algorithms such as **AES** before embedding it.
- Even if extracted, the data remains unreadable without the encryption key.

2. Randomized or Adaptive Embedding

- Hide data in randomly selected pixels using a secret key instead of sequential pixels.
- This makes steganalysis and extraction much more difficult

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understand ing of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technol ogy goes	Good understan ding with proper integration of most concepts	Basic understandi ng; limited or partial integration	Poor understandi ng; unable to integrate concepts
Experimen tal Design	30	Well-planned, systematic implementatio n with optimal solution	Correct implement ation with minor issues	Basic implementat ion with errors or inefficiencies	Incorrect or incomplete implement ation on
Use of Tools	20	Effective and advanced use of tools/technologie s with proper configuration	Appropriat e use of tools with correct execution	Limited or inconsisten t use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	interpretation n Basic analysis with limited	insights Poor or incorrect analysis of results
Documenta tion	10	Clear, well structured documentation with diagrams, code, and results	Organized documenta tion with necessary details	Basic documentat ion with missing details	Poor or incomplete documentat ion